
AP Physics C: Electricity and Magnetism

Practice Exam

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ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^8$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} (\text{N}\cdot\text{m}^2)/\text{kg}^2$
Universal gas constant, $R = 8.31 \text{ J}/(\text{mol}\cdot\text{K})$	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$	
1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$
Planck's constant,	$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s} = 4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$
	$hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m} = 1.24 \times 10^3 \text{ eV}\cdot\text{nm}$
Vacuum permittivity,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$
Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9 (\text{N}\cdot\text{m}^2)/\text{C}^2$	
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$
Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
1 atmosphere pressure,	$1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$

UNIT SYMBOLS	meter, m	mole, mol	watt, W	farad, F
	kilogram, kg	hertz, Hz	coulomb, C	tesla, T
	second, s	newton, N	volt, V	degree Celsius, °C
	ampere, A	pascal, Pa	ohm, Ω	electron volt, eV
	kelvin, K	joule, J	henry, H	

PREFIXES		
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan \theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- I. The frame of reference of any problem is inertial unless otherwise stated.
- II. The direction of current is the direction in which positive charges would drift.
- III. The electric potential is zero at an infinite distance from an isolated point charge.
- IV. All batteries and meters are ideal unless otherwise stated.
- V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.

ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

$$\vec{p} = m\vec{v}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_N|$$

$$\Delta E = W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2} m v^2$$

$$P = \frac{dE}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$\Delta U_g = mg \Delta h$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$I = \int r^2 dm = \sum m r^2$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$v = r \omega$$

$$\vec{L} = \vec{r} \times \vec{p} = I \vec{\omega}$$

$$K = \frac{1}{2} I \omega^2$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$a = \text{acceleration}$$

$$E = \text{energy}$$

$$F = \text{force}$$

$$f = \text{frequency}$$

$$h = \text{height}$$

$$I = \text{rotational inertia}$$

$$J = \text{impulse}$$

$$K = \text{kinetic energy}$$

$$k = \text{spring constant}$$

$$\ell = \text{length}$$

$$L = \text{angular momentum}$$

$$m = \text{mass}$$

$$P = \text{power}$$

$$p = \text{momentum}$$

$$r = \text{radius or distance}$$

$$T = \text{period}$$

$$t = \text{time}$$

$$U = \text{potential energy}$$

$$v = \text{velocity or speed}$$

$$W = \text{work done on a system}$$

$$x = \text{position}$$

$$\mu = \text{coefficient of friction}$$

$$\theta = \text{angle}$$

$$\tau = \text{torque}$$

$$\omega = \text{angular speed}$$

$$\alpha = \text{angular acceleration}$$

$$\phi = \text{phase angle}$$

$$\vec{F}_s = -k \Delta \vec{x}$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$x = x_{\max} \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_G| = \frac{G m_1 m_2}{r^2}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right|$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\kappa \epsilon_0 A}{d}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$I = \frac{dQ}{dt}$$

$$U_C = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

$$R = \frac{\rho \ell}{A}$$

$$\vec{E} = \rho \vec{J}$$

$$I = N e v_d A$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$P = I \Delta V$$

$$A = \text{area}$$

$$B = \text{magnetic field}$$

$$C = \text{capacitance}$$

$$d = \text{distance}$$

$$E = \text{electric field}$$

$$\mathcal{E} = \text{emf}$$

$$F = \text{force}$$

$$I = \text{current}$$

$$J = \text{current density}$$

$$L = \text{inductance}$$

$$\ell = \text{length}$$

$$n = \text{number of loops of wire per unit length}$$

$$N = \text{number of charge carriers per unit volume}$$

$$P = \text{power}$$

$$Q = \text{charge}$$

$$q = \text{point charge}$$

$$R = \text{resistance}$$

$$r = \text{radius or distance}$$

$$t = \text{time}$$

$$U = \text{potential or stored energy}$$

$$V = \text{electric potential}$$

$$v = \text{velocity or speed}$$

$$\rho = \text{resistivity}$$

$$\Phi = \text{flux}$$

$$\kappa = \text{dielectric constant}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{F} = \int I d\vec{\ell} \times \vec{B}$$

$$B_s = \mu_0 n I$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$U_L = \frac{1}{2} L I^2$$

ADVANCED PLACEMENT PHYSICS C EQUATIONS

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

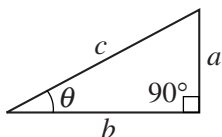
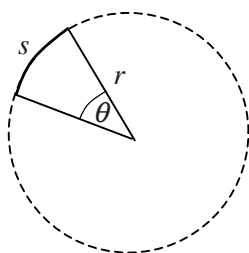
ℓ = length

w = width

r = radius

s = arc length

θ = angle



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

PHYSICS C: ELECTRICITY AND MAGNETISM

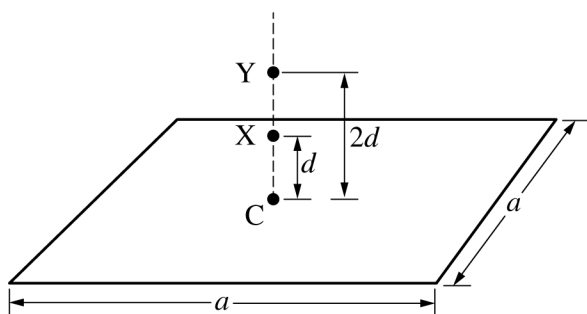
SECTION I

Time—45 minutes

35 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

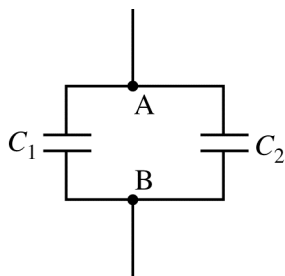
Questions 1-2



Note: Figure not drawn to scale.

The figure above shows a thin, square, nonconducting sheet of positive charge uniformly distributed over its area. The length of each side of the sheet is a . Point C is at the center of the sheet. Point X is a distance d above the center of the sheet, and point Y is a distance $2d$ above the center of the sheet. Assume $a \gg d$. The effect of gravity is negligible.

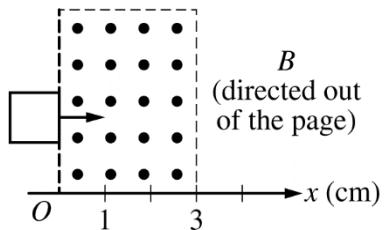
1. If the magnitude of the electric field at point X is E , what is the magnitude of the electric field at point Y ?
 (A) $E/4$
 (B) $E/2$
 (C) E
 (D) $2E$
 (E) $4E$
2. A positive point charge $+q$ is released from rest at point X . If the magnitude of the electric field at point X is E , what is the kinetic energy of the charge at point Y ?
 (A) qEd
 (B) $\sqrt{2}qEd$
 (C) $2qEd$
 (D) $2\sqrt{2}qEd$
 (E) $4qEd$



3. Capacitors C_1 and C_2 are connected as shown in the circuit above. The capacitance of C_1 is C , and the capacitance of C_2 is $C/3$. The equivalent capacitance between points A and B is C_{EQ} . A dielectric is inserted into capacitor C_2 , and the equivalent capacitance between points A and B is now $2C_{EQ}$. What is the value of the dielectric constant for this dielectric?

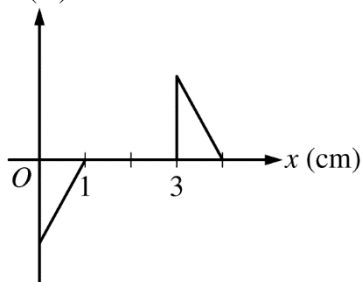
- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) The value cannot be determined without knowing the value of C .

4. Which of the following could be true for a Gaussian surface through which the net flux is zero?
- I. There are no charges inside the surface.
 - II. The net charge enclosed by the surface is zero.
 - III. The electric field is zero everywhere on the surface.
- (A) I only
 - (B) II only
 - (C) III only
 - (D) I and II only
 - (E) I, II, and III

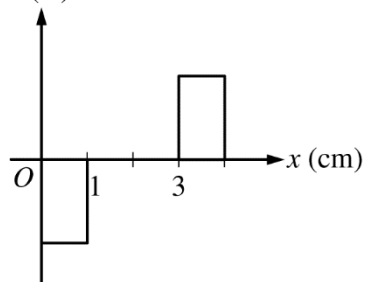


5. A square coil with sides of length 1.0 cm is moved through a region of uniform magnetic field at a constant speed, as shown in the figure above. Which of the following graphs best shows the current I in the coil as a function of the position x of the right edge as the coil moves through the the magnetic field, where counterclockwise current is positive?

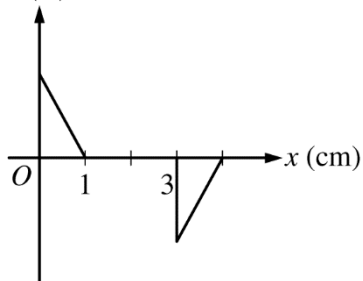
(A) I (A)



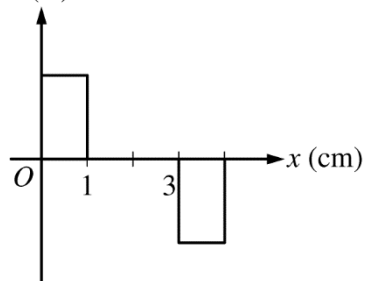
(B) I (A)



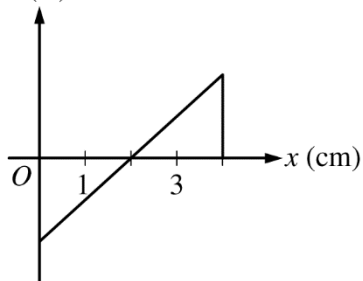
(C) I (A)

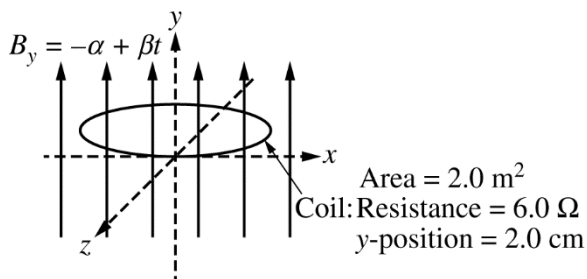


(D) I (A)

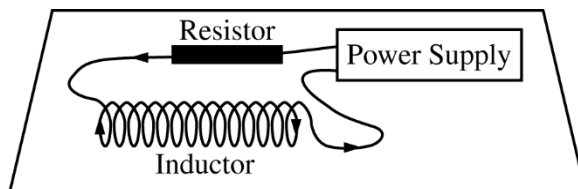


(E) I (A)



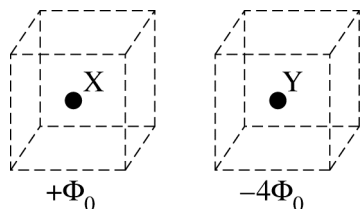


6. The y -component of the magnetic field B is given as a function of time t by the equation $B_y = -\alpha + \beta t$, where $\alpha = 4.0$ T and $\beta = 3.0$ T/s. A coil of wire with an area of 2.0 m^2 and a resistance of 6.0Ω is placed in this field, parallel to the xz -plane at $y = 2.0$ cm. The current in the coil at time $t = 1.2$ s is most nearly
- (A) 0
(B) 0.60 A
(C) 1.0 A
(D) 3.6 A
(E) 6.0 A

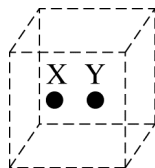


7. An inductor with inductance $L = 0.30$ H is connected in series with a resistor and both are connected to a power supply, as shown above. The power supply generates a current I that is given as a function of time t by the equation $I = I_0(1 - t/k)$, where $I_0 = 4.0$ A and $k = 2.0$ s. What is the magnitude of the potential difference across the inductor induced by the changing current?
- (A) 0.15 V
(B) 0.30 V
(C) 0.60 V
(D) 1.2 V
(E) 2.4 V
8. A parallel plate capacitor with air between the plates is charged and then disconnected from the source of emf. If the space between the plates is now filled with a dielectric, what happens to the capacitance of the capacitor and the magnitude of the electric field between the plates?
- | <u>Capacitance</u> | <u>Magnitude of the Electric Field</u> |
|--------------------|--|
| (A) Increases | Increases |
| (B) Increases | Remains constant |
| (C) Increases | Decreases |
| (D) Decreases | Increases |
| (E) Decreases | Decreases |

Questions 9-11

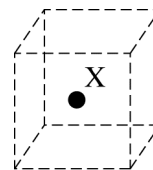


The figure above shows two charged spherical conductors, X and Y, which are equal in size. When each conductor is isolated and surrounded by a closed cubical surface, the total electric flux through the surfaces is $+\Phi_0$ for conductor X and $-4\Phi_0$ for conductor Y.



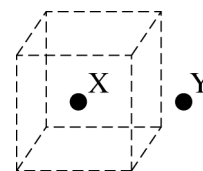
9. Conductor Y is brought into contact with conductor X and then separated. If the separation is small so that both conductors are inside the same closed cubical surface, as shown above, what is the total electric flux through the surface?

- (A) $-3\Phi_0$
- (B) $-\frac{5}{2}\Phi_0$
- (C) $-\frac{3}{2}\Phi_0$
- (D) $\frac{5}{2}\Phi_0$
- (E) It cannot be determined without knowing the distance separating the two conductors and the individual charges on each.

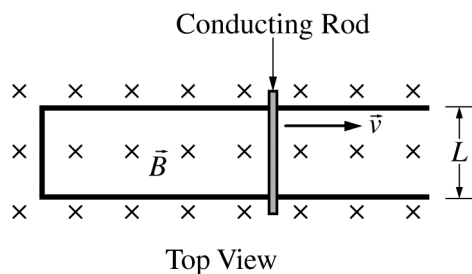


10. After being brought into contact with conductor X, conductor Y is moved a very large distance away from conductor X. What is the total electric flux through a closed cubical surface surrounding conductor X, as shown above?

- (A) $-3\Phi_0$
- (B) $-\frac{3}{2}\Phi_0$
- (C) $\frac{1}{2}\Phi_0$
- (D) Φ_0
- (E) $\frac{3}{2}\Phi_0$



11. Conductor Y is then moved closer to conductor X until it is just outside a closed cubical surface containing conductor X, as shown in the figure above. How would the total electric flux through the cubical surface change as conductor Y is moving?
- (A) It would increase.
 - (B) It would decrease.
 - (C) It would change sign.
 - (D) It would remain constant, with the flux through each side of the surface also remaining constant.
 - (E) It would remain constant, but the flux through each side of the cubical surface would change.



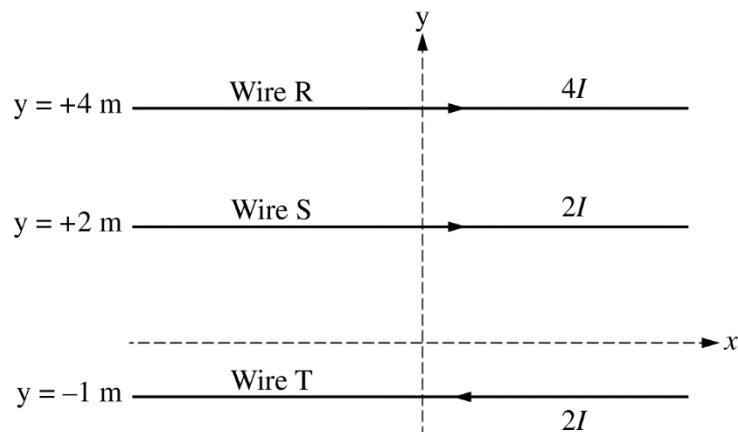
12. A conducting rod of resistance R is in electrical contact with a frictionless U-shaped rail of width L and negligible resistance. The rod is pulled to the right at a constant velocity $\vec{v}$. A magnetic field $\vec{B}$ is directed into the page, as shown in the figure above. Under these conditions, the electric power dissipated in the rod is P . If the velocity of the rod is doubled, the power dissipated in the rod is

(A) $P/4$
 (B) $P/2$
 (C) P
 (D) $2P$
 (E) $4P$

13. Which of the following will increase the magnetic field on the axis of a long solenoid?

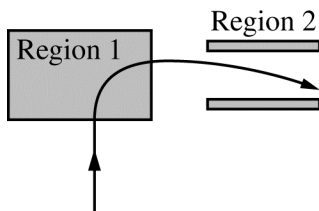
I. Increasing the current in the solenoid
 II. Increasing the cross-sectional area of the solenoid
 III. Inserting an iron core into the solenoid

(A) I only
 (B) III only
 (C) I and III only
 (D) II and III only
 (E) I, II, and III



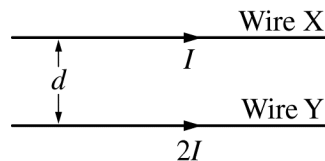
14. Three long wires, R, S, and T, are in the positions and carrying the currents indicated in the figure above. All currents are in the plane of the page and in the direction indicated. What are the magnitude and direction of the magnetic field at the origin?

	<u>Magnitude</u>	<u>Direction</u>
(A)	$\frac{\mu_0 I}{\pi}$	Into the page
(B)	$\frac{\mu_0 I}{\pi}$	Out of the page
(C)	$\frac{2\mu_0 I}{\pi}$	Into the page
(D)	$\frac{2\mu_0 I}{\pi}$	Out of the page
(E)	$\frac{4\mu_0 I}{\pi}$	Out of the page



15. A positively charged particle travels along the path shown in the figure above through region 1 and between two parallel plates in region 2. A magnetic field exists in region 1, and one of the plates in region 2 is at a higher potential than the other. What is the direction of the magnetic field in region 1, and which plate is at a higher potential in region 2?

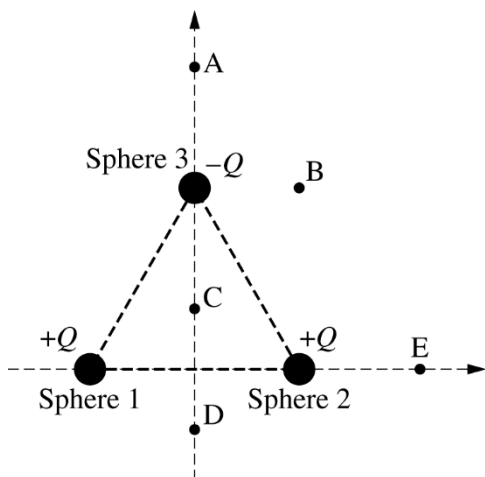
<u>Region 1</u>	<u>Plate at Higher Potential</u>
(A) Out of the page	Top plate
(B) Out of the page	Bottom plate
(C) Into the page	Top plate
(D) Into the page	Bottom plate
(E) To the right	Bottom plate



16. Two long, straight parallel wires X and Y are separated by a distance d and carry currents I and $2I$, as shown in the figure above. The force on wire X has magnitude F . If the current in each wire is both doubled and reversed in direction, which of the following is true of the magnitude and direction of the new force on wire X?

	<u>Magnitude</u>	<u>Direction</u>
(A)	F	Unchanged
(B)	$2F$	Reversed
(C)	$2F$	Unchanged
(D)	$4F$	Reversed
(E)	$4F$	Unchanged

Questions 17-18



Three identical spheres are equally spaced from one another. Spheres 1 and 2 have charge $+Q$, and sphere 3 has a charge of $-Q$, as shown in the figure above. Five different positions are labeled A, B, C, D, and E. Positions A, B, and C are all the same distance from sphere 3. Positions C, D, and E are all the same distance from sphere 2. All spheres and points lie in the same plane.

17. The electric potential is highest at position

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

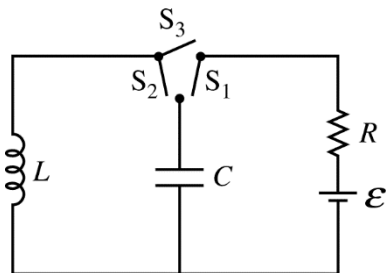
18. At which of the five positions are the horizontal and vertical components of the electric field directed toward the left and the top of the page, respectively?

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

19. The magnetic field at a perpendicular distance r from a long, straight current-carrying wire is directly proportional to

- (A) r
- (B) r^2
- (C) $1/r$
- (D) $1/r^2$
- (E) $\ln(r)$

Questions 20-21



A circuit is constructed using a battery of emf $\mathcal{E}$, a resistor of resistance R , a capacitor of capacitance C , an inductor of inductance L , and three switches, as shown in the figure above. The three switches are labeled S_1 , S_2 , and S_3 , and they can be operated independently.

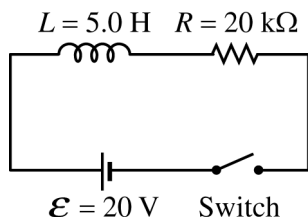
20. All switches are open, and there is no stored energy in the capacitor or the inductor. Switch S_3 is closed. What is the current in the inductor after steady state has been reached?

(A) $\frac{\mathcal{E}}{LR}$
 (B) $\frac{\mathcal{E}R}{L}$
 (C) $\frac{\mathcal{E}L}{R}$
 (D) $\frac{\mathcal{E}}{R}$
 (E) Zero

21. All switches are open, and there is no stored energy in the capacitor or the inductor. Switch S_1 is closed. After the capacitor is fully charged, switch S_1 is opened and switch S_2 is closed. Which of the following expressions represents the maximum current in the LC circuit?

(A) $\frac{\mathcal{E}}{R}$
 (B) $\mathcal{E}\sqrt{\frac{C}{L}}$
 (C) $\mathcal{E}\sqrt{\frac{L}{C}}$
 (D) $\frac{\mathcal{E}C}{L}$
 (E) $\frac{\mathcal{E}L}{C}$

Questions 22-23



After the switch is closed in the circuit above, the current in the circuit is given by $i = I(1 - e^{-t/\tau})$, where I and τ are constants.

22. What is the value of I ?

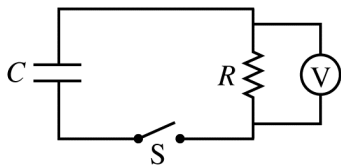
- (A) 0
- (B) 0.001 A
- (C) 0.80 A
- (D) 1.0 A
- (E) 4.0 A

23. What is the value of τ ?

- (A) $1.0 \times 10^{-5} \text{ s}$
- (B) $2.5 \times 10^{-4} \text{ s}$
- (C) 1.0 s
- (D) $4.0 \times 10^3 \text{ s}$
- (E) $1.0 \times 10^5 \text{ s}$

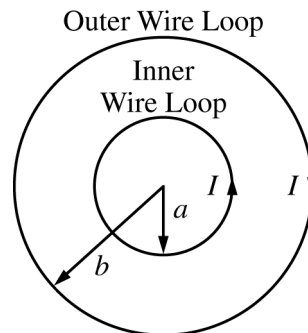
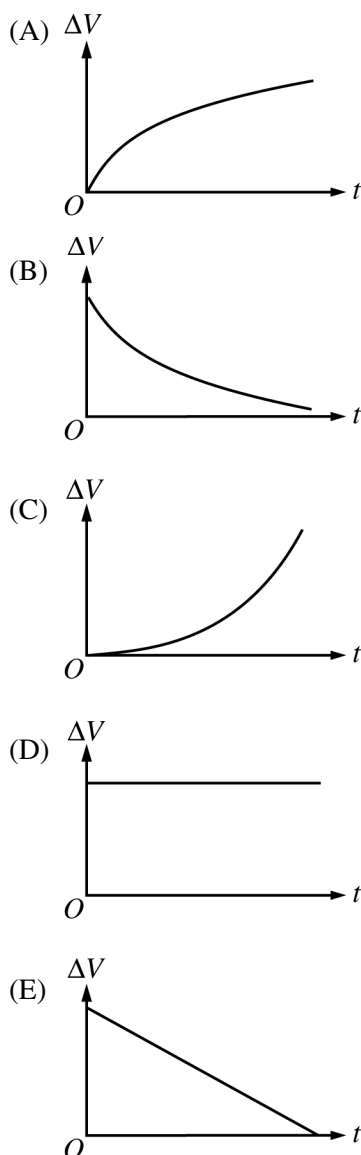
24. Two stationary point charges of unknown magnitude and sign are isolated from all other charges. If the electric field strength is zero at the midpoint of the line joining them, which of the following can be concluded about the charges?

- (A) They are equal in magnitude but opposite in sign.
- (B) They are equal in magnitude and have the same sign.
- (C) They are not necessarily equal in magnitude but have opposite signs.
- (D) They are not necessarily equal in magnitude but have the same sign.
- (E) None of the above can be concluded without additional information.



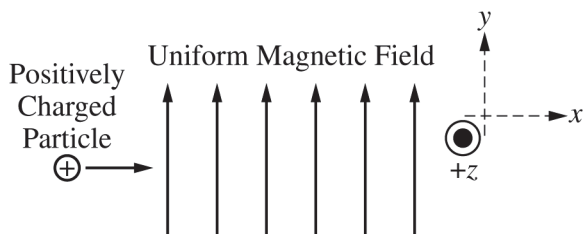
25. A charged capacitor C , resistor R , and switch S are connected in series. Voltmeter V is connected in parallel with the resistor, as shown in the figure above. At time $t = 0$, the switch is then closed.

Which of the following shows the potential difference ΔV reading on the voltmeter as a function of time t after the switch is closed?



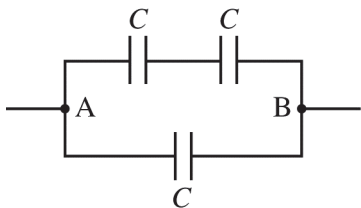
26. Two concentric circular wire loops carry equal currents I in opposite directions, as shown in the figure above. The inner loop has a radius a and carries a counterclockwise current. The outer loop has a radius b and carries a clockwise current. What is the magnitude and direction of the magnetic field at the center of the loops?

<u>Magnitude</u>	<u>Direction</u>
(A) $\frac{\mu_0 I}{2} \left(\frac{1}{a} - \frac{1}{b} \right)$	Out of the page
(B) $\frac{\mu_0 I}{2} \left(\frac{1}{a} - \frac{1}{b} \right)$	Into the page
(C) Zero	Undefined since magnitude is zero
(D) $\frac{\mu_0 I}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$	Out of the page
(E) $\frac{\mu_0 I}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$	Into the page



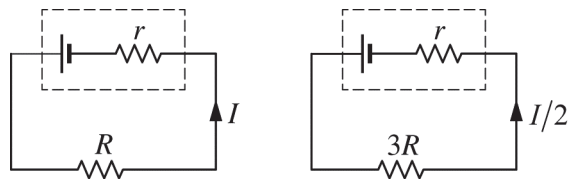
27. A positively charged particle moves in the positive x -direction in a uniform magnetic field directed in the positive y -direction. The net force on the particle could be zero if there is also an electric field present in the

(A) positive z -direction
 (B) negative z -direction
 (C) positive x -direction
 (D) negative x -direction
 (E) negative y -direction



28. Three identical capacitors with capacitance C are arranged as shown above. What is the equivalent capacitance between points A and B?

(A) $C/3$
 (B) $2C/3$
 (C) C
 (D) $3C/2$
 (E) $3C$



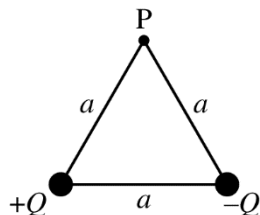
29. When a battery is connected to an external resistance R , as shown above on the left, there is a current I in the circuit. When the external resistance is changed to $3R$, the current changes to $I/2$, as shown above on the right. What is the internal resistance r of the battery?

(A) $R/4$
 (B) $R/2$
 (C) R
 (D) $2R$
 (E) $4R$

30. The advantage of using a dielectric in a capacitor is that it

(A) increases the capacitance of the capacitor
 (B) increases the conductivity of the capacitor
 (C) decreases the charging time of the capacitor
 (D) makes the capacitor dissipate less power
 (E) eliminates the electric field in the capacitor

Questions 31-33



Two small spheres have charges $+Q$ and $-Q$ and are located at the bottom corners of an equilateral triangle, as shown in the figure above. The equilateral triangle has sides of length a , and point P is at the top corner of the triangle.

31. The potential energy stored in the charge configuration shown is

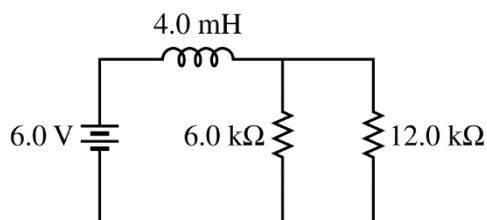
- (A) $\frac{Q^2}{4\pi\epsilon_0 a}$
- (B) $\frac{Q}{4\pi\epsilon_0 a^2}$
- (C) zero
- (D) $-\frac{Q}{4\pi\epsilon_0 a^2}$
- (E) $-\frac{Q^2}{4\pi\epsilon_0 a}$

32. The electric potential at point P due to the two spheres is

- (A) zero
- (B) $\frac{Q}{4\pi\epsilon_0 a}$
- (C) $\frac{Q}{4\pi\epsilon_0 a^2}$
- (D) $\frac{2Q}{4\pi\epsilon_0 a}$
- (E) $\frac{2Q}{4\pi\epsilon_0 a^2}$

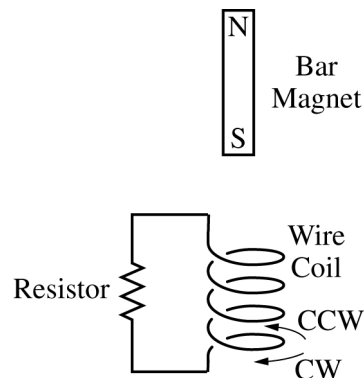
33. The direction of the electric field at point P due to the two spheres is

- (A) to the left
- (B) to the right
- (C) toward the bottom of the page
- (D) toward the top of the page
- (E) undefined, because the magnitude of the electric field at point P is zero



34. An inductor and two resistors are connected to an ideal battery, as shown in the figure above. What is the time constant for the circuit?

(A) $0.22 \mu\text{s}$
 (B) $0.33 \mu\text{s}$
 (C) $0.67 \mu\text{s}$
 (D) $1.0 \mu\text{s}$
 (E) $72 \mu\text{s}$



35. A bar magnet with its south pole pointing down is released from rest and falls through a wire coil, as shown above. A resistor is connected across the two ends of the coil. What current would be produced in the coil, as observed by a person directly above the coil?

(A) A clockwise current only
 (B) A counterclockwise current only
 (C) A current that is first clockwise and then counterclockwise
 (D) A current that is first counterclockwise and then clockwise
 (E) No current would be produced.

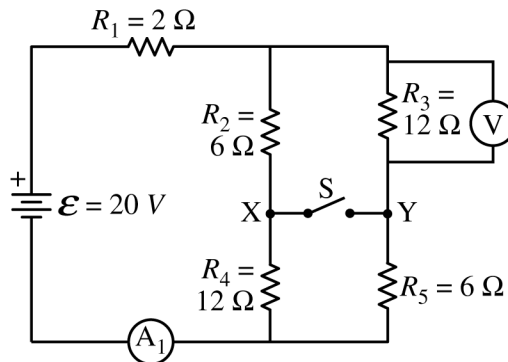
PHYSICS C: ELECTRICITY AND MAGNETISM

SECTION II

Time—45 minutes

3 Questions

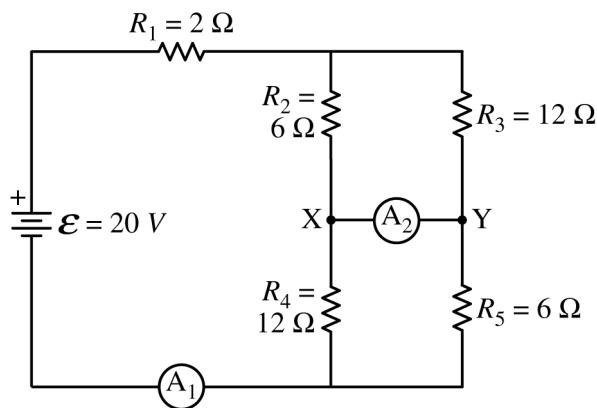
Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.



1. A circuit is created with an ideal battery and five resistors with the values shown in the figure above. An open switch is shown that can connect points X and Y. A voltmeter V and an ammeter A_1 are also shown in the figure.
 - (a) The switch is in the open position.
 - i. Calculate the current measured by the ammeter A_1 .
 - ii. Calculate the potential difference measured by the voltmeter V.
 - (b) The switch is now moved to the closed position.
 - i. Calculate the current measured by the ammeter A_1 .

- ii. Calculate the potential difference measured by the voltmeter V.

A student wants to determine if there is a current in the closed switch. The switch is now replaced with ammeter A_2 between points X and Y. The ammeter acts as a closed switch and can measure the current, if any, between the two points.



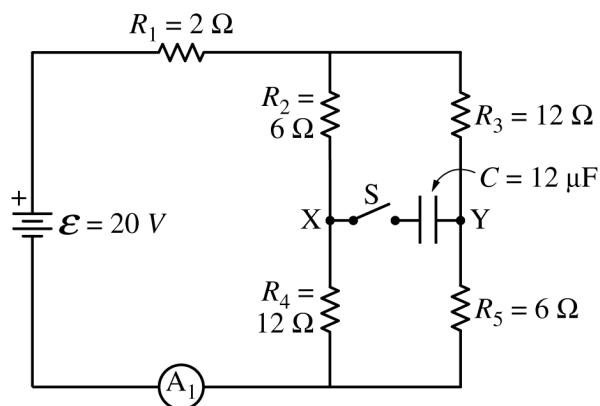
(c)

- i. Calculate the current measured by ammeter A_2 .

- ii. Indicate the direction of the current, if any, through the ammeter A_2 .

____ Left ____ Right ____ Undefined, because the current is zero

Question 1 continues on the next page.



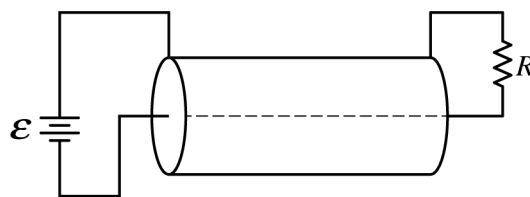
(d) The ammeter A_2 is replaced with switch S and capacitor $C = 12 \mu\text{F}$, as shown in the figure above. Switch S is closed.

i. Calculate the charge stored on the positive plate of capacitor C a long time after switch S is closed.

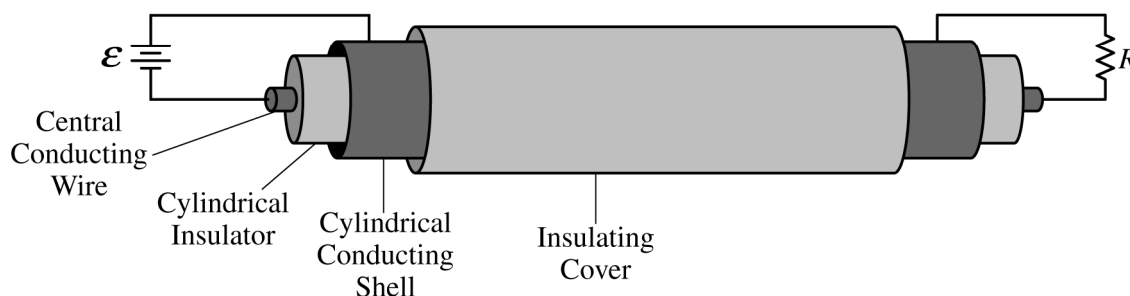
ii. Which point, X or Y , is at a higher electric potential?

____ X ____ Y

Justify your answer.



Schematic Diagram



Circuit Figure

2. A coaxial cable is composed of a central conducting wire surrounded by a cylindrical insulator, and around that is a conducting cylindrical shell with an insulating cover. A power supply of emf $\mathcal{E}$ is used at one end to connect the two conducting regions, and a resistive load of resistance R is connected at the other end of the cable to complete the circuit, as shown in the schematic diagram and circuit figure above.

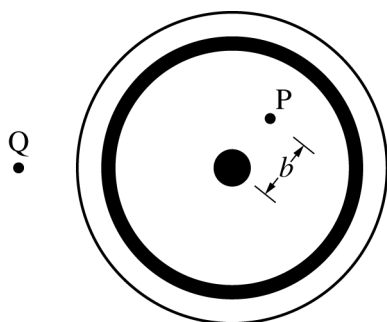


Figure 1

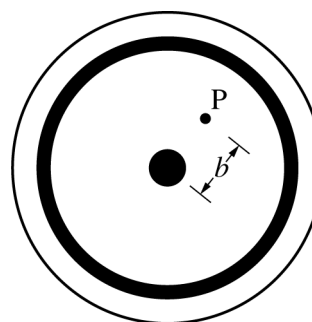


Figure 2

A cross section of the cable from the perspective of the power supply end is depicted above as if looking into the left end of the cable. Point P is inside the cylindrical insulator a distance b from the center of the cable. Point Q is outside the cable.

(a)

- On figure 1 shown above left, draw an Amperian loop that could be used to determine the magnitude of the magnetic field at point P in the diagram. Clearly mark this loop as “Loop 1.”
- On figure 1, draw an Amperian loop that could be used to determine the magnitude of the magnetic field at point Q in the diagram. Clearly mark this loop as “Loop 2.”
- On figure 2 shown above right, draw an arrow to show the direction of the magnetic field at point P. The arrow should start on and point away from the point in the direction of the magnetic field.

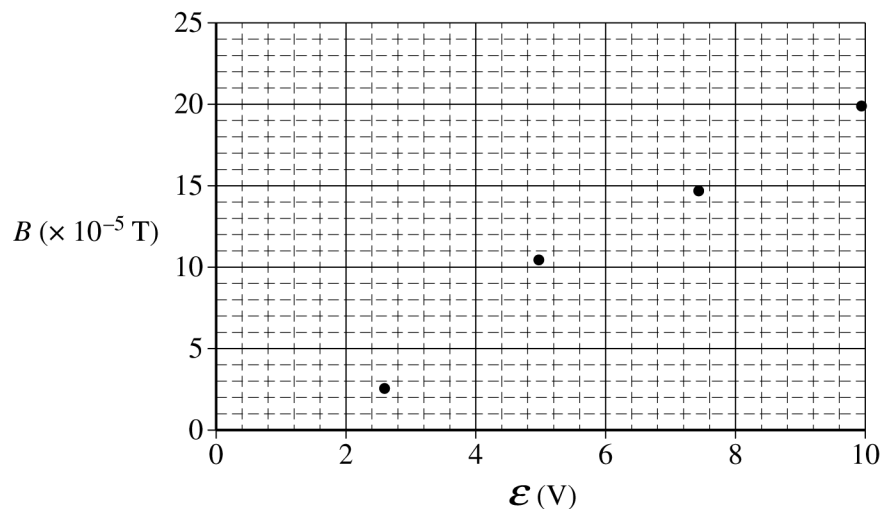
(b) Using Ampere's law, derive an expression for the magnitude of the magnetic field at point P. Express your answer in terms of ε , R , b , and physical constants, as appropriate.

(c) Determine the magnitude of the magnetic field at point Q.

Justify your answer.

Question 2 continues on the next page.

Some physics students conduct an experiment with a coaxial cable in which they connect the cable to a variable power supply and measure the resulting magnetic field strength at point P inside the cylindrical insulator using a magnetic field sensor. Point P is 0.0050 m from the center of the cable. The plot of the magnetic field strength B as a function of the emf $\mathcal{E}$ of the power supply is shown below.



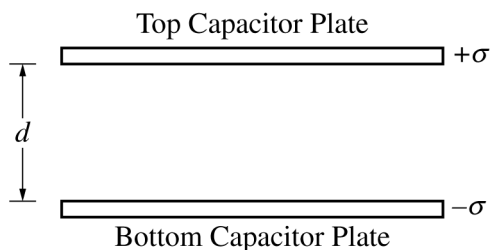
(d)

- Draw the best-fit line for the data.
- Use the straight line to calculate the resistance R of the resistor used in the experiment.

- (e) The lab group uses a multimeter to measure the resistance of the resistor and determines that their experimental value for R from part (d) is approximately 10% larger than that found with the multimeter. Describe a physical reason that might account for this result.

- i. Describe a physical reason for the vertical axis intercept.

GO ON TO THE NEXT PAGE.



Note: Figure not drawn to scale.

3. A parallel plate capacitor is connected to a battery, fully charged, and then isolated from the battery. The plates are given equal and opposite charge densities σ , and the separation between the plates is d , as shown in the figure above. The area of each plate has a value of A .
- (a) Derive an expression for the energy stored in the capacitor. Express your answer in terms of σ , d , A , and physical constants, as appropriate.



While the capacitor is isolated from the battery, an uncharged metal slab of thickness x and area A is inserted between the capacitor plates so that it is an equal distance from the two plates, as shown in the figure above.

- (b) On the figure above, draw an arrow to indicate the direction of the electric field in the following three regions of space between the capacitor plates. If the electric field is zero in any region, indicate this by writing “ $E = 0$ ” in that region.
- The region between the top plate and the metal slab
 - The region inside the metal slab
 - The region between the metal slab and the bottom plate
- (c) On the lines below, rank the regions according to the magnitude of the electric field, with 1 being the largest magnitude. If any regions have the same magnitude of electric field, give them the same ranking.
- ____ The region between the top plate and the metal slab
- ____ The region inside the metal slab
- ____ The region between the metal slab and the bottom plate

- (d) Is the capacitance of the capacitor with the metal slab greater than, less than, or equal to the capacitance of the original capacitor without the slab?

_____ Greater than _____ Less than _____ Equal to

Justify your answer.

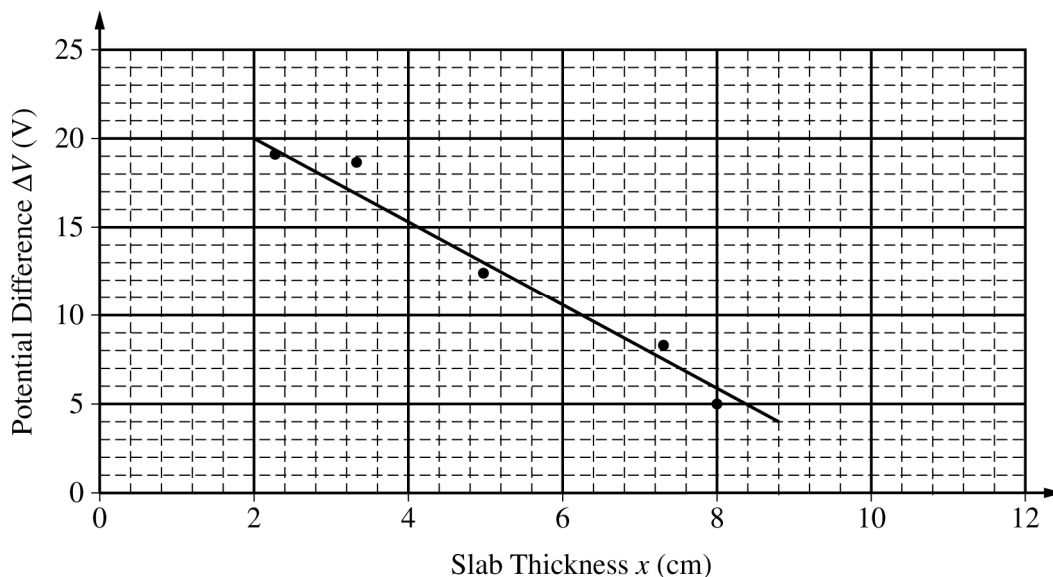
- (e) When charged to the original charge density, is the energy stored in the capacitor with the metal slab greater than, less than, or equal to the original energy stored in the capacitor calculated in part (a) ?

_____ Greater than _____ Less than _____ Equal to

Justify your answer.

Question 3 continues on the next page.

Some physics students conduct an experiment in which they charge the capacitor to the same potential difference, isolate it from the battery, insert the metal slab, and measure the new potential difference across the capacitor. This experiment is repeated with slabs of different thickness. The plot of the potential difference ΔV across the capacitor as a function of the thickness x of the slab is shown below. A best-fit line for the data has been drawn.



(f) Use the best-fit line to calculate the following:

i. σ , the charge density on the plates

ii. d , the distance between the capacitor plates

**Answer Key for AP Physics C: Electricity and Magnetism
Practice Exam, Section I**

Question 1: C

Question 2: A

Question 3: D

Question 4: E

Question 5: B

Question 6: C

Question 7: C

Question 8: C

Question 9: A

Question 10: B

Question 11: E

Question 12: E

Question 13: C

Question 14: C

Question 15: A

Question 16: E

Question 17: D

Question 18: B

Question 19: C

Question 20: D

Question 21: B

Question 22: B

Question 23: B

Question 24: B

Question 25: B

Question 26: A

Question 27: B

Question 28: D

Question 29: C

Question 30: A

Question 31: E

Question 32: A

Question 33: B

Question 34: D

Question 35: C

2018 AP Physics C: Electricity and Magnetism Question Descriptors and Performance Data

Multiple-Choice Questions

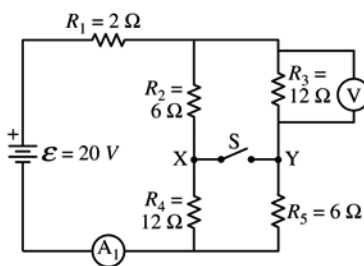
Questions	Topic	Key	% Correct
1	Electrostatics	C	29
2	Electrostatics	A	40
3	Conductors/capacitors/dielectrics	D	40
4	Electrostatics	E	64
5	Electromagnetism	B	40
6	Electromagnetism	C	51
7	Electromagnetism	C	69
8	Conductors/capacitors/dielectrics	C	50
9	Electrostatics	A	79
10	Electrostatics	B	65
11	Electrostatics	E	49
12	Electromagnetism	E	46
13	Electromagnetism	C	43
14	Magnetostatics	C	59
15	Magnetostatics	A	65
16	Magnetostatics	E	39
17	Electrostatics	D	53
18	Electrostatics	B	76
19	Magnetostatics	C	62
20	Electromagnetism	D	65
21	Electromagnetism	B	24
22	Electromagnetism	B	58
23	Electromagnetism	B	59
24	Electrostatics	B	69
25	Electric circuits	B	71
26	Magnetostatics	A	19
27	Magnetostatics	B	63
28	Electric circuits	D	73
29	Electric circuits	C	62
30	Conductors/capacitors/dielectrics	A	86
31	Electrostatics	E	43
32	Electrostatics	A	50
33	Electrostatics	B	74
34	Electromagnetism	D	61
35	Electromagnetism	C	42

AP[®] PHYSICS C: ELECTRICITY AND MAGNETISM
2018 SCORING GUIDELINES

Question 1

15 points total

**Distribution
of points**



A circuit is created with an ideal battery and five resistors with the values shown in the figure above. An open switch is shown that can connect points X and Y. A voltmeter V and an ammeter A_1 are also shown in the figure.

(a) The switch is in the open position.

i) 3 points

Calculate the current measured by the ammeter A_1 .

For correct equivalent resistance of the circuit	1 point
$R_{24} = R_2 + R_4 = 6\ \Omega + 12\ \Omega = 18\ \Omega$	
$R_{35} = R_3 + R_5 = 12\ \Omega + 6\ \Omega = 18\ \Omega$	
$\frac{1}{R_P} = \frac{1}{R_{24}} + \frac{1}{R_{35}} = \frac{1}{18\ \Omega} + \frac{1}{18\ \Omega} = \frac{2}{18\ \Omega} = \frac{1}{9\ \Omega} \therefore R_P = 9\ \Omega$	
$R_{EQ} = R_1 + R_P = 2\ \Omega + 9\ \Omega = 11\ \Omega$	
For consistent substitution into a correct equation (i.e., Ohm's law or Kirchhoff's law)	1 point
$I = \frac{V}{R} = \frac{(20\ \text{V})}{(11\ \Omega)}$	
For the correct answer with the units	1 point
$I = 1.82\ \text{A}$	

ii) 1 point

Calculate the potential difference measured by the voltmeter V.

For correct substitution (consistent with a(i)) into a correct equation to calculate the potential difference across R_3	1 point
$V_P = \mathcal{E} - V_1 = \mathcal{E} - IR_1 = (20\ \text{V}) - (1.82\ \text{A})(2\ \Omega) = 16.4\ \text{V}$	
$I_{35} = \frac{V_P}{R_{35}} = \frac{(16.4\ \text{V})}{(18\ \Omega)} = 0.91\ \text{A}$	
$V_3 = I_{35}R_3 = (0.91\ \text{A})(12\ \Omega) = 10.9\ \text{V}$	

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Question 1 (continued)

**Distribution
of points**

(b) The switch is now moved to the closed position.

i) 3 points

Calculate the current measured by the ammeter A_1 .

For the correct equivalent resistance of the circuit		1 point
$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{6\ \Omega} + \frac{1}{12\ \Omega} = \frac{3}{12\ \Omega} = \frac{1}{4\ \Omega} \therefore R_{23} = 4\ \Omega$		
$\frac{1}{R_{45}} = \frac{1}{R_4} + \frac{1}{R_5} = \frac{1}{12\ \Omega} + \frac{1}{6\ \Omega} = \frac{3}{12\ \Omega} = \frac{1}{4\ \Omega} \therefore R_{45} = 4\ \Omega$		
$R_S = R_{23} + R_{45} = 4\ \Omega + 4\ \Omega = 8\ \Omega$		
$R_{EQ} = R_1 + R_S = 2\ \Omega + 8\ \Omega = 10\ \Omega$		
For consistent substitution into a correct equation (i.e., Ohm's law or Kirchhoff's law)		1 point
$I = \frac{V}{R} = \frac{(20\ \text{V})}{(10\ \Omega)}$		
For the correct answer with the units		1 point
$I = 2.0\ \text{A}$		

ii) 1 points

Calculate the potential difference measured by the voltmeter V.

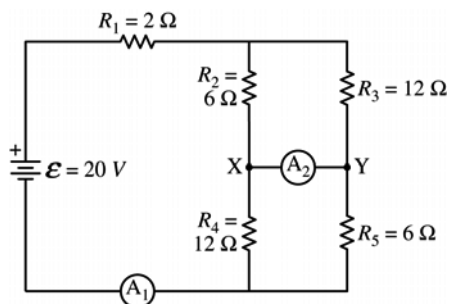
For correct substitution (consistent with b(i)) into a correct equation to calculate the potential difference across R_3		1 point
$V_{23} = IR_{23} = (2.0\ \text{A})(4\ \Omega) = 8\ \text{V}$		

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2018 SCORING GUIDELINES

Question 1 (continued)

**Distribution
of points**

A student wants to determine if there is a current in the closed switch. The switch is now replaced with ammeter A_2 between points X and Y. The ammeter acts as a closed switch and can measure the current, if any, between the two points.



- (c)
i) 2 points

Calculate the current measured by ammeter A_2 .

For using a correct equation to calculate the current through R_2 and R_4 (or R_3 and R_5)	1 point
$I_2 = \frac{V_{23}}{R_2} = \frac{(8 \text{ V})}{(6 \Omega)} = 1.33 \text{ A}$	
$I_4 = \frac{V_{45}}{R_4} = \frac{\mathcal{E} - V_1 - V_{23}}{R_4} = \frac{\mathcal{E} - IR_1 - V_{23}}{R_4} = \frac{(20 \text{ V}) - (2 \text{ A})(2 \Omega) - (8 \text{ V})}{(12 \Omega)} = 0.67 \text{ A}$	
For indicating that the current through the ammeter is the difference between the currents through the two resistors	1 point
$I_A = I_2 - I_4 = (1.33 \text{ A}) - (0.67 \text{ A}) = 0.67 \text{ A}$	

- ii) 1 point

Indicate the direction of the current, if any, through the ammeter A_2 .

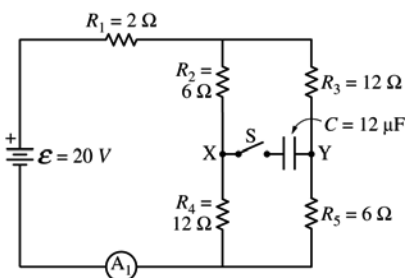
____ Left ____ Right ____ Undefined, because the current is zero

For selecting “Right” Note: If a student calculated I_A in Part c(i) to be zero then the consistent answer of “Undefined” earns 1 point.	1 point
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Question 1 (continued)

**Distribution
of points**



- (d) The ammeter A_2 is replaced with switch S and capacitor $C = 12 \mu\text{F}$, as shown in the figure above. Switch S is closed.

i) 3 points

Calculate the charge stored on the positive plate of capacitor C a long time after switch S is closed.

For calculating the potential at points X and Y	1 point
$I_{24} = \frac{V_P}{R_{24}} = \frac{(16.4 \text{ V})}{(18 \Omega)} = 0.91 \text{ A}$	
$V_2 = I_{24}R_2 = (0.91 \text{ A})(6 \Omega) = 5.5 \text{ V}$	
$V_X = \mathcal{E} - V_1 - V_2 = (20 \text{ V}) - (1.82 \text{ A})(2 \Omega) - (5.5 \text{ V}) = 10.9 \text{ V}$	
$V_Y = \mathcal{E} - V_1 - V_3 = (20 \text{ V}) - (1.82 \text{ A})(2 \Omega) - (10.9 \text{ V}) = 5.5 \text{ V}$	
For indicating that the potential difference across the capacitor is the difference in potential at points X & Y	1 point
$V_C = V_X - V_Y = (10.9 \text{ V}) - (5.5 \text{ V}) = 5.4 \text{ V}$	
For substituting into the equation for the charge stored on the capacitor	1 point
$Q = CV = (12 \mu\text{F})(5.4 \text{ V}) = 65.3 \mu\text{C}$	

ii) 1 point

Which point, X or Y, is at a higher electric potential?

____ X ____ Y

Justify your answer.

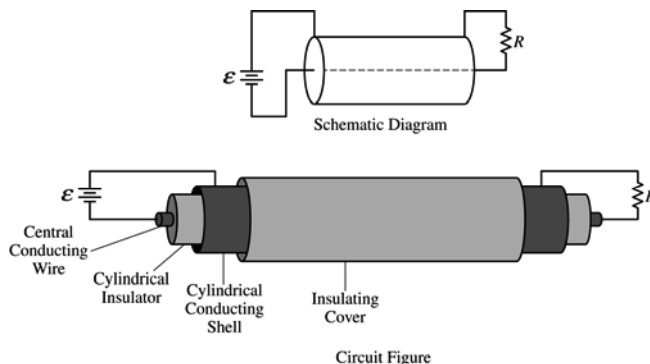
Selecting “X”	
For a correct justification	1 point
Example: The point X is at higher potential because the current flows to the right. The current flows from high potential to low potential. Example: The potential at X is 10.9 V and the potential at Y is 5.5 V. So point X is at the higher potential.	

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Question 2

15 points total

**Distribution
of points**



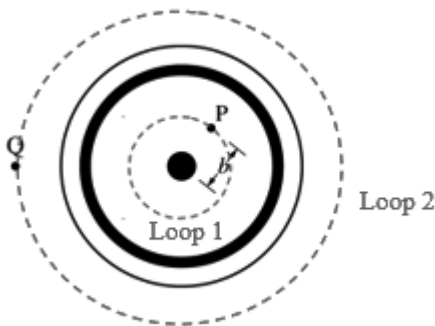
A coaxial cable is composed of a central conducting wire surrounded by a cylindrical insulator, and around that is a conducting cylindrical shell with an insulating cover. A power supply of emf $\mathcal{E}$ is used at one end to connect the two conducting regions, and a resistive load of resistance R is connected at the other end of the cable to complete the circuit, as shown in the schematic diagram and circuit figure above.

A cross section of the cable from the perspective of the power supply end is depicted above as if looking into the left end of the cable. Point P is inside the cylindrical insulator a distance b from the center of the cable. Point Q is outside the cable.

(a)

i) 1 point

On figure 1 shown above left, draw an Amperian loop that could be used to determine the magnitude of the magnetic field at point P in the diagram. Clearly mark this loop as “Loop 1.”



For properly drawing and labeling a circle concentric with the other circles and through point P	1 point
--	---------

ii) 1 point

On figure 1, draw an Amperian loop that could be used to determine the magnitude of the magnetic field at point Q in the diagram. Clearly mark this loop as “Loop 2.”

For properly drawing and labeling a circle concentric with the other circles and through point Q	1 point
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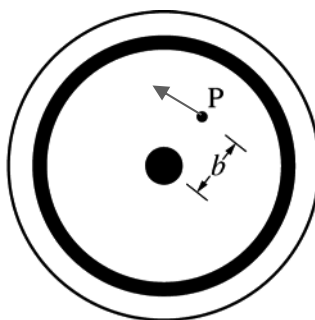
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Question 2 (continued)

**Distribution
of points**

- (a) Con't
 iii) 2 points

On figure 2 shown above right, draw an arrow to show the direction of the magnetic field at point P. The arrow should start on and point away from the point in the direction of the magnetic field.



For properly drawing a straight arrow at point P that indicates the magnetic field is tangent to Loop 1 from part (a)(i)	1 point
For indicating a counter-clockwise magnetic field around Loop 1 from part (a)(i)	1 point

- (b) 3 points

Using Ampere's law, derive an expression for the magnitude of the magnetic field at point P. Express your answer in terms of $\mathcal{E}$, R , b , and physical constants, as appropriate.

For using Ampere's law at point P	1 point
$\oint \vec{B} d\vec{\ell} = \mu_0 I$	
For correctly substituting for $\oint d\vec{\ell}$	1 point
$B(2\pi b) = \mu_0 I$	
For correctly substituting for I	1 point
$B(2\pi b) = \mu_0 \frac{\mathcal{E}}{R} \therefore B = \frac{\mu_0 \mathcal{E}}{2\pi b R}$	

- (c) 2 points

Determine the magnitude of the magnetic field at point Q.
 Justify your answer.

For indicating that the magnetic field at point Q is zero	1 point
For a correct justification	1 point
<i>Example: Since Loop 2 encloses both the inside wire and the outer conducting shell and the two of them have equal and opposite currents, the net enclosed current is zero. Thus the magnetic field at point Q is zero.</i>	

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Question 2 (continued)

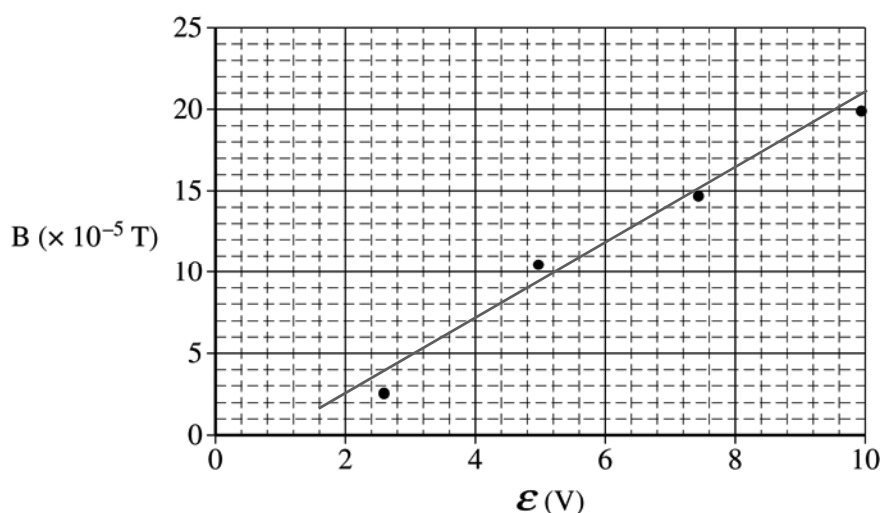
**Distribution
of points**

Some physics students conduct an experiment with a coaxial cable in which they connect the cable to a variable power supply and measure the resulting magnetic field strength at point P inside the cylindrical insulator using a magnetic field sensor. Point P is 0.0050 m from the center of the cable. The plot of the magnetic field strength B as a function of the emf $\mathcal{E}$ of the power supply is shown below.

(d)

i) 1 point

Draw the best-fit line for the data.



For properly drawing a trend line on the graph	1 point
--	---------

ii) 2 points

Use the straight line to calculate the resistance R of the resistor used in the experiment.

For correctly calculating the slope from the best-fit line and not the data points	1 point
$\text{slope} = \frac{\Delta B}{\Delta \mathcal{E}} = \frac{(20 - 5)(\text{T} \times 10^{-5})}{(9.4 - 3.2)(\text{V})} = 2.42 \times 10^{-5} \text{ T/V}$	
For correctly relating the slope to the equation from part (b)	1 point
$\text{slope} = \frac{\mu_0}{2\pi b R} \therefore R = \frac{\mu_0}{\text{slope} \times 2\pi b} = \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})}{(2.42 \times 10^{-5} \text{ T/V})(2\pi)(0.0050 \text{ m})} = 1.65 \Omega$	

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Question 2 (continued)

**Distribution
of points**

(e) 1 point

The lab group uses a multimeter to measure the resistance of the resistor and determines that their experimental value for R from part (d) is approximately 10% larger than that found with the multimeter. Describe a physical reason that might account for this result.

For an appropriate explanation for the difference in the resistance values		1 point
<i>Example: There is resistance in the wires, thereby increasing the total experimental resistance.</i>		

(f) The lab group notices that when the current is reversed in the cable and the experiment is again performed, the plot has a positive vertical axis intercept equal in magnitude to the negative vertical axis intercept in the plot shown before part (d).

i) 1 point

Describe a physical reason for the vertical axis intercept.

For an appropriate explanation for the vertical axis intercept		1 point
<i>Example: The magnetic field sensor was not zeroed at the start of the experiment to account for the Earth's magnetic field</i>		

ii) 1 point

Describe a physical reason that the vertical axis intercept switches from negative to positive when the current in the cable is reversed.

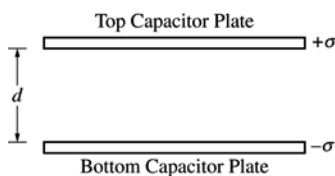
For an appropriate explanation for the intercept switching from negative to positive		1 point
<i>Example: When the current is reversed, the cable's magnetic field switches direction but the external magnetic field from the Earth remains constant.</i>		

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Question 3

15 points total

**Distribution
of points**



Note: Figure not drawn to scale.

A parallel plate capacitor is connected to a battery, fully charged, and then isolated from the battery. The plates are given equal and opposite charge densities σ , and the separation between the plates is d , as shown in the figure above. The area of each plate has a value of A .

(a) 3 points

Derive an expression for the energy stored in the capacitor. Express your answer in terms of σ , d , A , and physical constants, as appropriate.

For using an appropriate equation for calculating the energy stored in a capacitor		1 point
$U = \frac{q^2}{2C}$		
For substituting correctly into above equation		1 point
$U = \frac{(\sigma A)^2}{2 \frac{\epsilon_0 A}{d}}$		
For a correct answer		1 point
$U = \frac{dA\sigma^2}{2\epsilon_0}$		

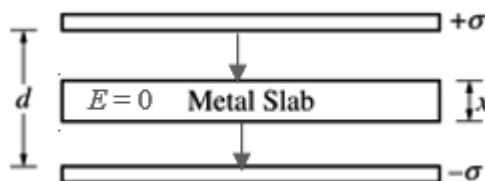
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Question 3 (continued)

**Distribution
of points**

While the capacitor is isolated from the battery, an uncharged metal slab of thickness x and area A is inserted between the capacitor plates so that it is an equal distance from the two plates, as shown in the figure above.

- (b) On the figure above, draw an arrow to indicate the direction of the electric field in the following three regions of space between the capacitor plates. If the electric field is zero in any region, indicate this by writing “ $E = 0$ ” in that region.



- i) 1 point

The region between the top plate and the metal slab

For properly drawing an arrow between the top plate and the metal slab pointing downward	1 point
--	---------

- ii) 1 point

The region inside the metal slab

For indicating that $E = 0$ inside the metal slab	1 point
---	---------

- iii) 1 point

The region between the metal slab and the bottom plate

For properly drawing an arrow between the metal slab and the bottom plate pointing downward	1 point
---	---------

- (c) 1 point

On the lines below, rank the regions according to the magnitude of the electric field, with 1 being the largest magnitude. If any regions have the same magnitude of electric field, give them the same ranking.

_____ The region between the top plate and the metal slab

_____ The region inside the metal slab

_____ The region between the metal slab and the bottom plate

For ranking both regions between a plate and the metal slab as 1 and the region inside the metal slab as 2	1 point
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Question 3 (continued)

**Distribution
of points**

(d) 2 points

Is the capacitance of the capacitor with the metal slab greater than, less than, or equal to the capacitance of the original capacitor without the slab?

____ Greater than ____ Less than ____ Equal to

Justify your answer.

For selecting “Greater than” with an attempt at a relevant justification		1 point
For a correct justification		1 point
<i>Example: Inserting the metal slab effectively decreases the distance over which the electric field exists in the capacitor. Since the distance between the plates is effectively smaller, the capacitance is increased.</i>		
<i>Example: Inserting the metal slab decreases the extent of the electric field between the plates, which lowers the electric potential. If the electric potential between plates of a capacitor is lowered and the charge remains constant, the capacitance is increased.</i>		

(e) 2 points

When charged to the original charge density, is the energy stored in the capacitor with the metal slab greater than, less than, or equal to the original energy stored in the capacitor calculated in part (a) ?

____ Greater than ____ Less than ____ Equal to

Justify your answer.

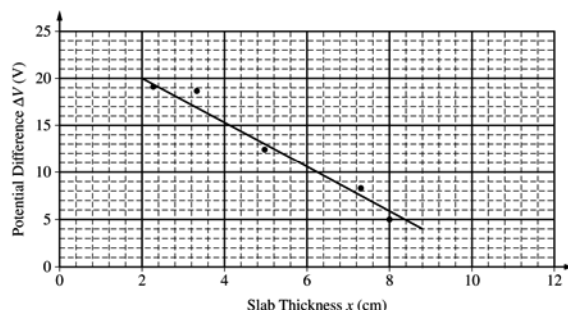
For selecting “Less than” with an attempt at a relevant justification or for selecting an answer consistent with part (d)		1 point
For a correct justification		1 point
<i>Example: Since the equivalent capacitance with the slab is greater than the original capacitance, and the energy stored is related to capacitance by $U = q^2 / (2C)$, the energy stored on the capacitor will be less with the metal slab.</i>		

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Question 3 (continued)

**Distribution
of points**

Some physics students conduct an experiment in which they charge the capacitor to the same potential difference, isolate it from the battery, insert the metal slab, and measure the new potential difference across the capacitor. This experiment is repeated with slabs of different thickness. The plot of the potential difference ΔV across the capacitor as a function of the thickness x of the slab is shown below. A best-fit line for the data has been drawn.



(f) Use the best-fit line to calculate the following:

i) 2 points

σ , the charge density on the plate

For correctly calculating the slope using appropriate values from the best-fit line		1 point
$\text{slope} = \frac{\Delta V}{\Delta x} = \frac{(15 - 5)(\text{V})}{(4.1 - 8.4)(\times 10^{-2} \text{ m})} = -232 \text{ V/m}$		
For correctly relating the slope to the charge density on the plates		1 point
$\text{slope} = -\frac{\sigma}{\epsilon_0} \therefore \sigma = -\text{slope} \times \epsilon_0 = -(-232 \text{ V/m})(8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2))$		
$\sigma = 2.06 \times 10^{-9} \text{ C/m}^2$		

ii) 2 points

d , the distance between the capacitor plates

For appropriately using the best fit line		1 point
$V = \frac{\sigma(d - x)}{\epsilon_0} \therefore (d - x) = \frac{V\epsilon_0}{\sigma} \therefore d = x + \frac{V\epsilon_0}{\sigma}$		
For an answer, with units, consistent with the charge density found above		1 point
$d = 0.02 \text{ m} + \frac{(20 \text{ V})(8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2))}{(2.06 \times 10^{-9} \text{ C/m}^2)}$		
$d = 10.6 \text{ cm}$		

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Question 3 (continued)

**Distribution
of points**

- (f) con't
 ii) con't

<i>Alternate solution</i>		<i>Alternate points</i>
<i>For using the extension of the best fit line to the x-axis</i>		<i>1 point</i>
<i>For an answer, with units, consistent with the extension of the line</i>		<i>1 point</i>
<i>d = 10.6 cm</i>		